

# PROJECT REPORT

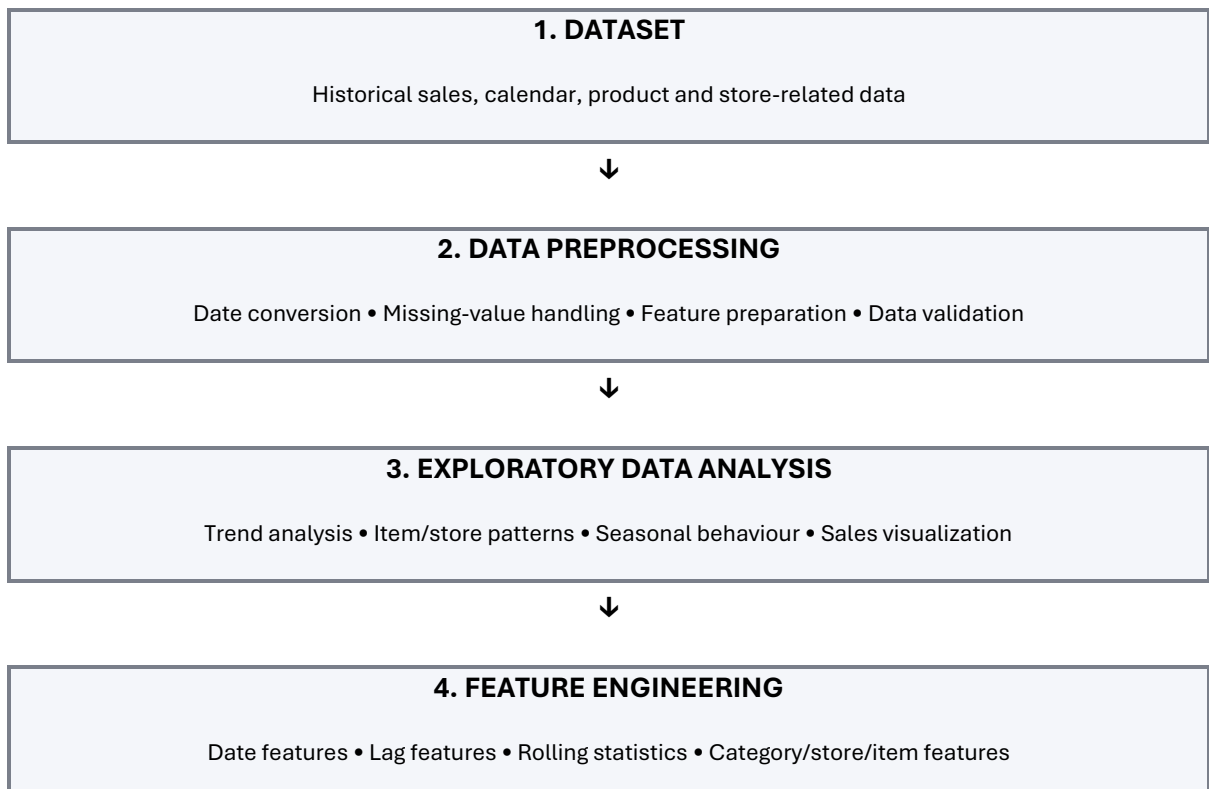
# FORESIGHT

## AI-Powered Demand & Inventory Intelligence Platform

Machine Learning • Python • Streamlit • Power BI  
*Project Documentation*

### System Architecture

The system follows a sequential data-to-insight architecture. Each stage produces the output required by the next stage. The architecture below shows the complete flow used in Project FORESIGHT.





## 5. MACHINE LEARNING MODEL

Random Forest Regression • Train/validation split • Model training • Prediction



## 6. FORECAST & INVENTORY ANALYSIS

Demand prediction • Stockout risk • Overstock risk • Inventory recommendations



## 7. STREAMLIT APPLICATION

Interactive prediction • Filters • KPIs • Charts • Forecast and inventory insights



## 8. POWER BI DASHBOARD

Sales trends • KPIs • Product/store analysis • Time analysis • Business insights



## 9. BUSINESS OUTPUT

Demand planning • Better stock decisions • Reduced stockout/overstock risk

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Demand planning • Better stock decisions • Reduced stockout/overstock risk

### 1. Project Overview

**Project FORESIGHT – AI-Powered Demand & Inventory Intelligence Platform** is a machine learning-based solution designed to forecast product demand and support better inventory planning.

The project uses historical sales data to identify **sales trends, seasonal patterns, product-level demand and store-level demand**. The data is processed and transformed into useful features, which are then used to train a **Random Forest Regression model** for demand prediction.

The predicted demand is further used to identify potential **stockout and overstock risks** and provide inventory-related insights. The results are presented through an interactive **Streamlit application** and a **Power BI dashboard**, making the analysis easier to understand and use for business decision-making.

#### **Main purpose:**

To convert historical sales data into **demand forecasts and actionable inventory insights**, helping businesses improve stock planning, reduce unnecessary inventory and minimize the risk of insufficient stock.

### 1. Problem Statement

Businesses need to maintain sufficient inventory to meet customer demand while avoiding unnecessary excess stock. Manual analysis of historical sales data makes it difficult to identify demand patterns, seasonal changes, and product-level variations.

**Project FORESIGHT** addresses this problem by analysing historical sales data and using machine learning to forecast future demand. The predicted

demand can then be used to identify potential **stockout and overstock risks** and support better inventory planning and decision-making.

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## 2. Objectives

The main objectives of Project FORESIGHT are:

- To analyse historical sales data and understand demand patterns.
  - To identify sales trends and seasonal variations.
  - To perform data preprocessing and feature engineering.
  - To create useful forecasting features such as date, lag, and rolling features.
  - To develop a **Random Forest Regression** model for demand prediction.
  - To generate future demand predictions.
  - To identify potential stockout and overstock risks.
  - To create an interactive **Streamlit application** for displaying predictions and insights.
  - To develop a **Power BI dashboard** for business analysis and visualization.
  - To provide useful insights for better inventory and stock planning.
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## 3. Dataset Details

The project uses a **historical sales dataset** containing time-based sales information along with product/item and store-related information.

The dataset is used to analyse historical demand and train the machine learning model.

### Main data components

- **Date** – Used to analyse time-based demand patterns.
- **Store** – Represents different store locations.
- **Item/Product** – Represents individual products.
- **Sales** – Represents the quantity of products sold.

## Dataset Processing

The dataset goes through the following steps:

1. Data loading and inspection
2. Data type verification
3. Date conversion
4. Missing-value checking and handling
5. Feature preparation
6. Exploratory data analysis
7. Feature engineering
8. Model training and prediction

The prepared data is then used for **demand forecasting and inventory analysis**.

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## 4. Technology Stack

| Technology | Purpose                        |
|------------|--------------------------------|
| Python     | Main programming language      |
| Pandas     | Data manipulation and analysis |
| NumPy      | Numerical operations           |
| Matplotlib | Data visualization             |

| Technology               | Purpose                                |
|--------------------------|----------------------------------------|
| Scikit-learn             | Machine learning                       |
| Random Forest Regression | Demand forecasting                     |
| Jupyter Notebook         | Data analysis and model development    |
| Streamlit                | Interactive web application            |
| Power BI                 | Business dashboard and reporting       |
| Pickle (.pkl)            | Saving and loading the trained model   |
| Git & GitHub             | Version control and project management |

### Overall Technology Flow

**Dataset → Python/Pandas → Data Preprocessing → Feature Engineering → Random Forest Model → Demand Prediction → Streamlit + Power BI → Business Insights**

### Complete Project Workflow

Project FORESIGHT follows an end-to-end machine learning and business intelligence workflow.

#### Step 1: Dataset Loading

The historical sales dataset is loaded into the Python environment using Pandas. The structure, columns, data types and records are checked before further processing.

#### Step 2: Data Preprocessing

The dataset is prepared for analysis and modelling. This includes:

- Checking data types

- Converting the date column into a proper date format
- Checking missing values
- Handling required data inconsistencies
- Preparing the required columns

### **Step 3: Exploratory Data Analysis**

The historical sales data is analysed to understand:

- Sales trends
- Daily and monthly patterns
- Product/item demand
- Store-level demand
- High- and low-demand periods

### **Step 4: Feature Engineering**

Additional features are created from the available data to improve demand prediction. These include:

- Date-based features
- Day
- Month
- Year
- Lag-based features
- Rolling statistics
- Item/store-related features

### **Step 5: Machine Learning**

The prepared dataset is used to train a **Random Forest Regression** model. The model learns patterns from historical sales data and generates demand predictions.

### **Step 6: Demand Prediction**

The trained model is used to predict demand for the required records. The prediction results are stored for further analysis.

### **Step 7: Inventory Analysis**

The predicted demand is analysed to identify possible:

- Stockout risk
- Overstock risk
- High-demand products
- Low-demand products

### **Step 8: Streamlit Application**

The trained model and processed data are integrated into an interactive Streamlit application. Users can view KPIs, predictions, charts and inventory-related insights.

### **Step 9: Power BI Dashboard**

The project data is connected to Power BI to create an interactive business dashboard containing KPIs, trends, filters and visualizations.

### **Overall Workflow**

**Dataset → Data Preprocessing → EDA → Feature Engineering → Random Forest Model → Demand Prediction → Inventory Analysis → Streamlit → Power BI → Business Insights**

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### **Random Forest Model Details**

#### **Model Used**

## Random Forest Regression

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to produce a prediction.

For this project, Random Forest Regression is used because the target value, **sales/demand**, is a numerical value.

### Working of the Model

1. Historical sales data is provided to the model.
2. The engineered features are used as input variables.
3. Multiple decision trees are created.
4. Each tree produces a prediction.
5. The predictions from the trees are combined.
6. The final output represents the predicted demand.

### Input Features

The model uses prepared features such as:

- Date-related features
- Item/product information
- Store information
- Historical/lag demand
- Rolling statistics
- Other relevant engineered features

### Model Output

The output of the model is the **predicted demand/sales value**, which is subsequently used for forecasting and inventory analysis.

### Model Saving

After training, the model is saved as a **.pkl file**. This allows the trained model to be loaded directly into the Streamlit application without training the model every time.

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## **Prediction & Inventory Analysis**

The main purpose of the prediction stage is to estimate product demand based on patterns learned from historical data.

### **Prediction Process**

**Prepared Data → Trained Random Forest Model → Predicted Demand → Inventory Analysis**

The predicted demand is compared with the available business information to understand inventory requirements.

### **Inventory Analysis**

The system can be used to identify:

#### **Stockout Risk**

When expected demand is higher than the available stock or planned inventory level.

#### **Overstock Risk**

When available inventory is significantly higher than expected demand.

#### **High-Demand Products**

Products showing comparatively high predicted or historical demand.

#### **Low-Demand Products**

Products showing comparatively low predicted or historical demand.

### **Business Benefit**

The analysis can help businesses:

- Plan inventory according to expected demand
- Reduce unnecessary stock

- Minimize potential stock shortages
- Identify products requiring attention
- Make data-driven inventory decisions

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## 5+ Visualizations and KPIs

The project includes visualizations to understand historical sales and predicted demand.

### Purpose:

To understand expected future demand based on the trained model.

### Additional Visualizations

The project can also include:

- Monthly sales analysis
- Yearly sales analysis
- Demand distribution
- Item-level analysis
- Store-level analysis

### Main KPIs

| KPI                  | Purpose                    |
|----------------------|----------------------------|
| Total Sales          | Shows overall sales volume |
| Average Sales        | Shows average demand       |
| Total Items/Products | Shows product coverage     |
| Number of Stores     | Shows store coverage       |
| Predicted Demand     | Shows forecasted demand    |

| KPI                         | Purpose                            |
|-----------------------------|------------------------------------|
| <b>Stockout Risk Count</b>  | Shows potential shortage cases     |
| <b>Overstock Risk Count</b> | Shows potential excess-stock cases |

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## 5. Streamlit Application

The Streamlit application provides an interactive interface for the Project FORESIGHT forecasting system.

### Main Functions

The application includes:

- Project overview
- Dataset information
- KPIs
- Sales visualizations
- Demand prediction
- Product/item analysis
- Store analysis
- Inventory insights
- Interactive filters

### Application Workflow

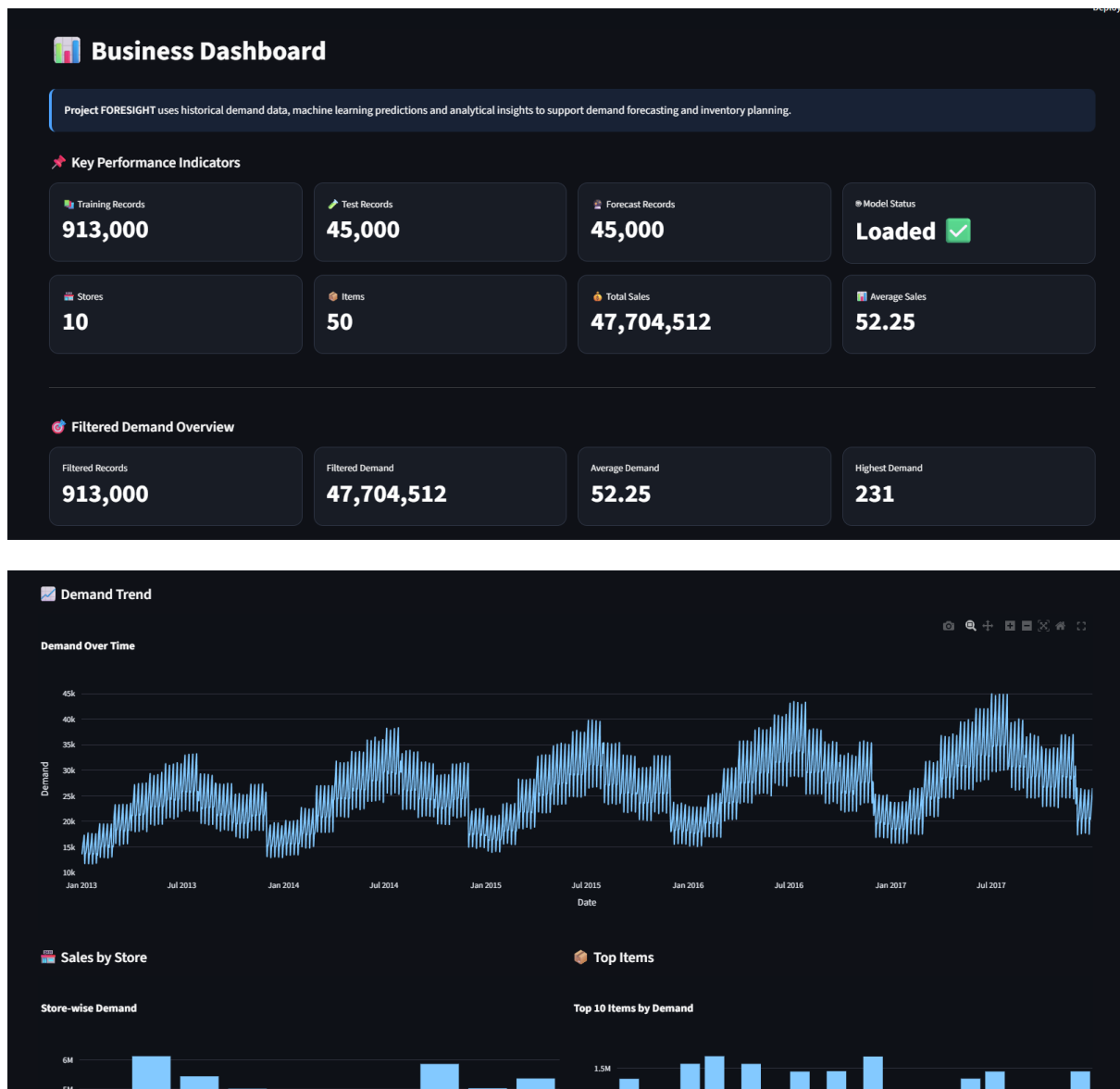
**User Input/Selection → Processed Data → Trained .pkl Model → Prediction → Visualization → Inventory Insight**

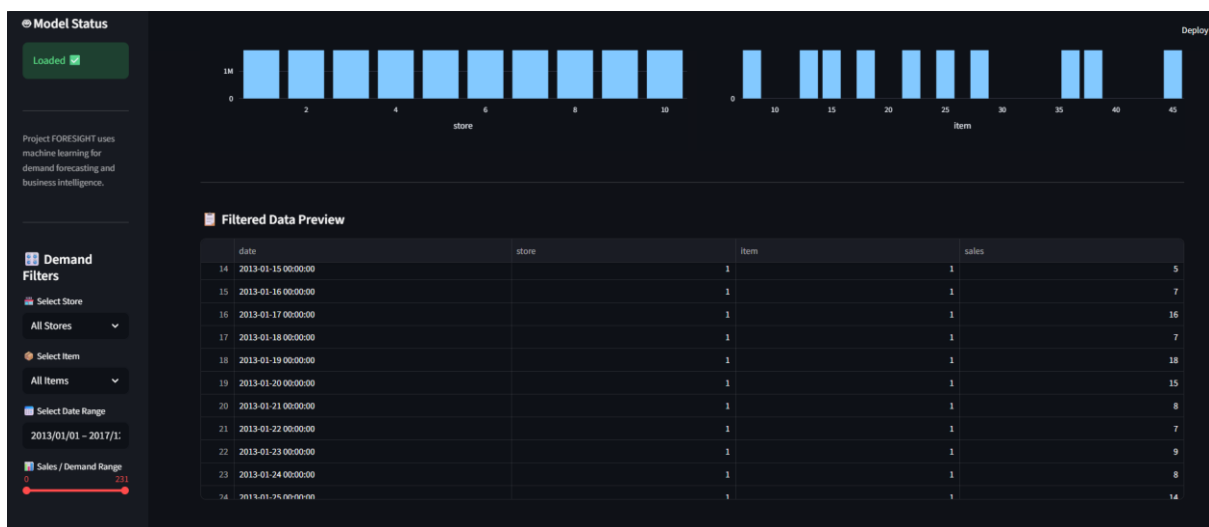
The saved machine learning model is loaded into the application so that predictions can be generated without retraining the model.

### Purpose

The Streamlit application converts the technical machine learning model into a simple and interactive interface that can be used by non-technical users.

## Screenshot





## 6. Power BI Dashboard

Power BI is used to create a business intelligence dashboard for Project FORESIGHT.

The dashboard provides an interactive view of historical sales, product performance, store performance and demand-related information.

### Main Dashboard Components

#### KPI Cards

- Total Sales
- Average Sales

- Total Items
- Number of Stores
- Predicted Demand

### **Visualizations**

- Sales trend over time
- Sales by product/item
- Sales by store
- Day-wise sales
- Monthly/yearly analysis
- Demand analysis

### **Filters/Slicers**

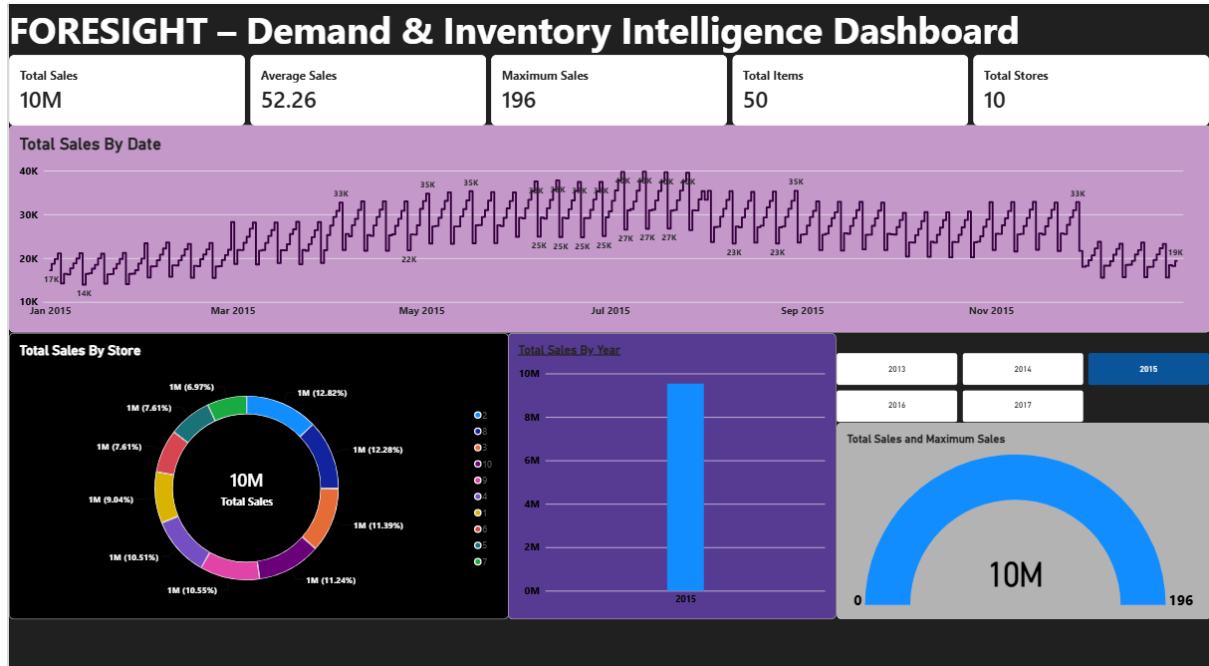
The dashboard can provide filters such as:

- Date
- Year
- Month
- Store
- Item/Product

### **Purpose**

Power BI provides a business-friendly way to explore the project data and identify important trends and patterns.

### **Screenshot**



## 7. Project Outputs / Files

The major outputs generated during the project are:

### Data Files

- Original dataset
- Processed/cleaned dataset
- Prediction dataset

### Machine Learning Files

- Trained Random Forest model
- Model file: .pkl

### Analysis Files

- Jupyter Notebook
- Exploratory analysis
- Generated visualizations

## Application Files

- Streamlit application
- requirements.txt

## Dashboard Files

- Power BI dashboard
- Power BI project/report file

## Documentation

- Project README
- Project Report

## Repository

The complete project can be maintained using **Git and GitHub** for version control and project submission.

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## 8. Challenges & Solutions

| Challenge                                          | Solution                                                      |
|----------------------------------------------------|---------------------------------------------------------------|
| Historical sales data contains time-based patterns | Created date-based features and performed time-based analysis |
| Raw data required preparation before modelling     | Performed data preprocessing and validation                   |
| Demand changes between products/stores             | Included product/item and store-related features              |
| Historical demand alone is difficult to interpret  | Used visualizations and exploratory analysis                  |

| Challenge                                                        | Solution                                          |
|------------------------------------------------------------------|---------------------------------------------------|
| Machine learning output can be difficult for users to understand | Developed Streamlit application                   |
| Business users need an easy reporting interface                  | Created Power BI dashboard                        |
| Model needs to be reused without retraining                      | Saved the trained model as a .pkl file            |
| Inventory decisions require demand information                   | Used predicted demand for inventory-risk analysis |

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## 9. Conclusion

**Project FORESIGHT – AI-Powered Demand & Inventory Intelligence Platform** demonstrates an end-to-end machine learning and business intelligence solution for demand forecasting and inventory planning.

The project starts with historical sales data and applies **data preprocessing, exploratory data analysis, feature engineering and Random Forest Regression** to generate demand predictions.

The predictions are then used to support inventory analysis and identify potential stockout and overstock situations.

The project combines **Python, Machine Learning, Streamlit and Power BI** to transform raw sales data into meaningful visualizations, predictions and business insights.

Overall, Project FORESIGHT provides a structured approach for using historical sales information to support **data-driven demand forecasting and inventory planning**.

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## 10. Future Scope

The project can be further improved with the following features:

### **1. Advanced Forecasting Models**

Additional algorithms such as:

- XGBoost
- LightGBM
- ARIMA
- Prophet
- LSTM

can be evaluated and compared.

### **2. Automated Model Retraining**

The system can automatically retrain the model whenever new sales data becomes available.

### **3. Real-Time Inventory Data**

Real-time stock levels can be integrated with the forecasting system.

### **4. Automated Alerts**

The application can send alerts when:

- Stockout risk becomes high
- Overstock risk becomes high
- Demand increases significantly

### **5. Improved Forecast Accuracy**

Hyperparameter tuning and advanced feature engineering can be used to improve model performance.

### **6. Cloud Deployment**

The Streamlit application can be deployed online so that it can be accessed from anywhere.

## 7. Automated Reporting

Power BI reports can be connected to updated data sources for automated business reporting.

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## 11. Project Folder Structure

Use a structure like this in your report:

PROJECT FORESIGHT/

|

└─ Dataset/

|

└─ train.csv

|

└─ test.csv

|

└─ Notebook/

|

└─ Project\_FORESIGHT.ipynb

|

└─ Model/

|

└─ trained\_model.pkl

|

└─ Predictions/

|

└─ predictions.csv

|

└─ Visualizations/

```
|  ├── sales_trend.png
|  ├── day_wise_sales.png
|  ├── top_items.png
|  ├── store_analysis.png
|  └── demand_prediction.png
|
|  ├── Streamlit/
|  |   ├── app.py
|  |   └── requirements.txt
|  |
|  |   ├── PowerBI/
|  |   |   └── Project_FORESIGHT.pbix
|  |   |
|  |   └── README.md
|  |
|  └── Project_Report.pdf
```

**Important:** Change the filenames above if your actual GitHub folder has different names. The structure should match what you have actually uploaded.

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## 12. Final Submission Checklist

Before submitting Project FORESIGHT, check the following:

### Project Files

- ☐ Dataset included
- ☐ Jupyter Notebook included

- ☐ Data preprocessing included
- ☐ Feature engineering included
- ☐ Random Forest model included
- ☐ .pkl model saved
- ☐ Prediction file generated
- ☐ Visualizations included

### **Streamlit**

- ☐ Streamlit application working
- ☐ KPIs displayed
- ☐ Charts displayed
- ☐ Prediction functionality working
- ☐ .pkl model connected
- ☐ requirements.txt included
- ☐ Live Streamlit link added if required

### **Power BI**

- ☐ Power BI file completed
- ☐ KPI cards created
- ☐ Date relationship configured
- ☐ Charts created
- ☐ Slicers/filters added
- ☐ Dashboard is readable

### **GitHub**

- ☐ Project uploaded to GitHub
- ☐ README added
- ☐ Folder structure organized
- ☐ Large/unnecessary files handled correctly
- ☐ Streamlit live link added to README if available

### **Project Report**

- ☐ Project Overview
- ☐ Problem Statement
- ☐ Objectives

- ☐ Dataset Details
  - ☐ Technology Stack
  - ☐ System Architecture
  - ☐ Complete Workflow
  - ☐ Random Forest Model
  - ☐ Prediction & Inventory Analysis
  - ☐ Visualizations & KPIs
  - ☐ Streamlit Application
  - ☐ Power BI Dashboard
  - ☐ Challenges & Solutions
  - ☐ Conclusion
  - ☐ Future Scope
  - ☐ Project Folder Structure
  - ☐ Actual screenshots inserted
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## **13. Dedicated Screenshot Section**

### **13.1 Streamlit Application**

**Figure: Project FORESIGHT Streamlit Application**

**INSERT YOUR ACTUAL STREAMLIT SCREENSHOT HERE**

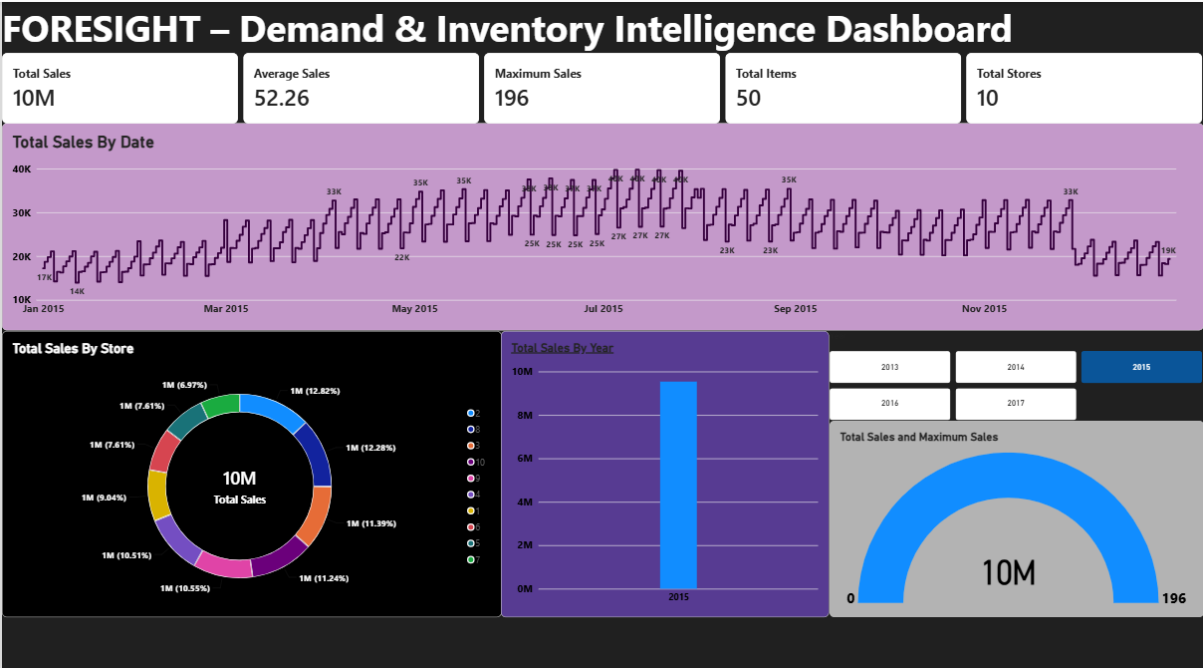
**Description:**

The Streamlit application provides an interactive interface for viewing project KPIs, sales analysis, demand predictions and inventory-related insights.

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### **13.2 Power BI Dashboard**

Figure: Project FORESIGHT Power BI Dashboard



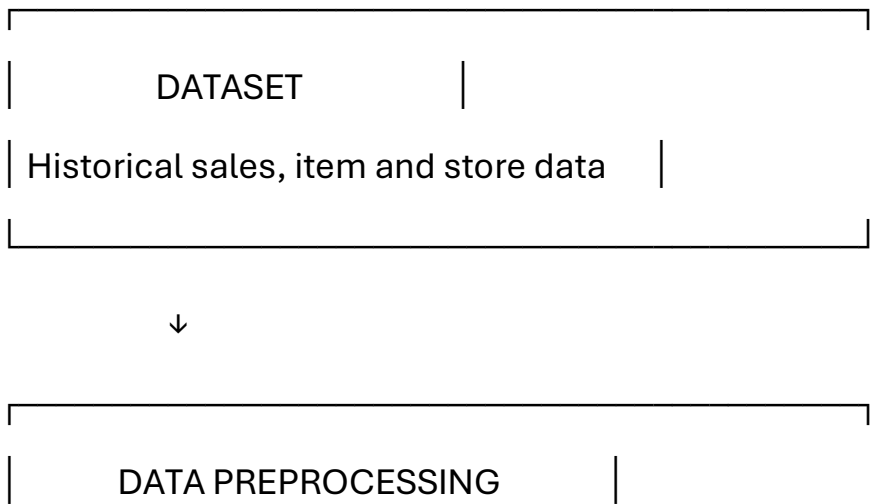
INSERT YOUR ACTUAL POWER BI SCREENSHOT HERE

Description:

The Power BI dashboard provides an interactive business intelligence view of sales performance, demand patterns, KPIs and product/store-level analysis.

Final Architecture for the Report

Put this as the **System Architecture** diagram in your report:



| Date conversion • Missing values • Validation |



| EXPLORATORY DATA ANALYSIS (EDA) |

| Trends • Patterns • Item • Store analysis |



| FEATURE ENGINEERING |

| Date • Lag • Rolling • Item/Store features |



| RANDOM FOREST REGRESSION |

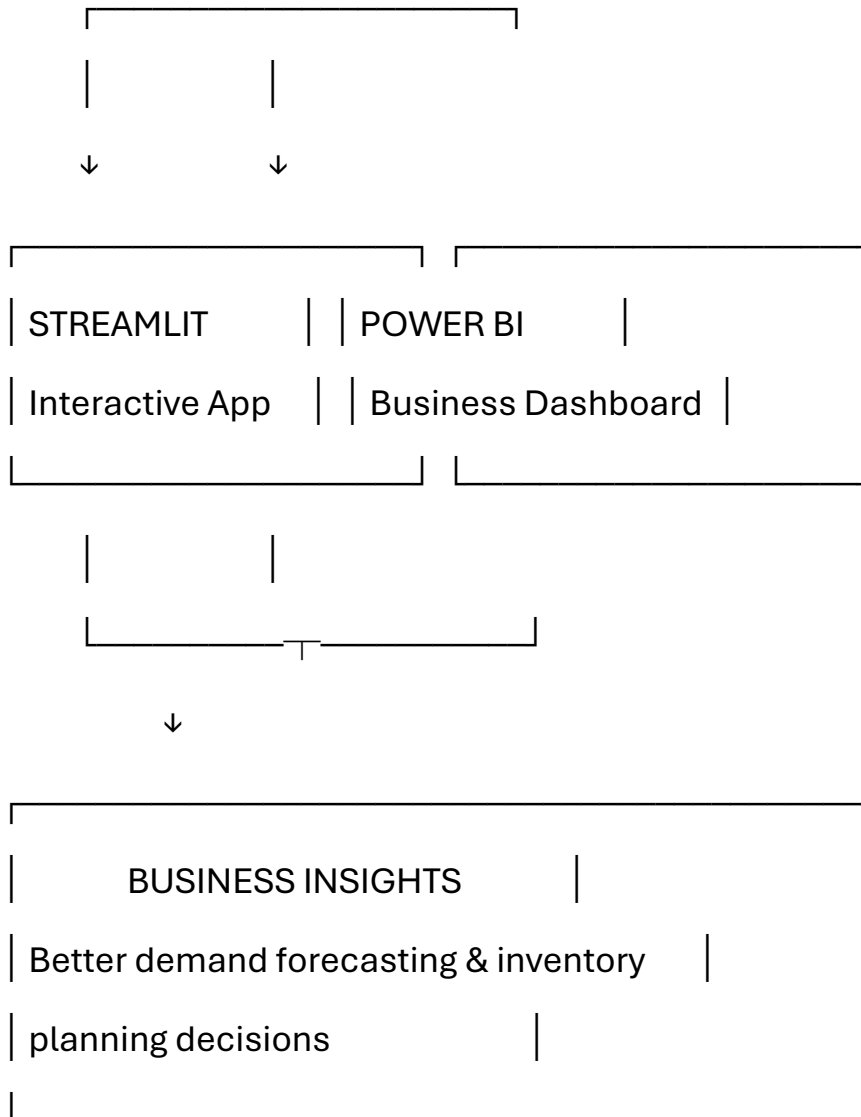
| Demand Forecasting |



| PREDICTION & INVENTORY ANALYSIS |

| Demand • Stockout Risk • Overstock Risk |





This is the structure I recommend putting directly into your **final project report**.